Computer Algorithms and Complexity

1. Verify that the constructed 5 x 5 magic square satisfies the magic square properties by showing that the sums of all rows, columns, and both diagonals are equal to the magic constant.
2. Analyze the given code segment and determine the count of each type of statement:

```
for (int i = 0; i < 4; i++) {
   for (int j = 0; j < 4; j++) {
      cout << "i=" << i << ", j=" << j << endl;
      cout << "i=" << i << ", j=" << j << endl;
   }
}
```

3. Explain the sequence of moves required to solve the Tower of Hanoi problem with 3 disks. Use a source peg (A), an auxiliary peg (B), and a destination peg (C). Illustrate each step involved in the process.
4. Write the pseudo code for Magic Square.
5. Describe the step-by-step process of the Selection algorithm for sorting an array [8, 4, 6, 9, 2, 3, 1] in ascending order.
6. Write down the pseudocode of magic square for odd order logic and generate an N x N magic square step by step where N=3.
7. Describe the step-by-step process of the Insertion algorithm for sorting an array [8, 4, 6, 9, 2, 3, 1] in ascending order.
8. Explain the sequence of moves required to solve the Tower of Hanoi problem with 3 disks. Use a source peg (A), an auxiliary peg (B), and a destination peg (C). Illustrate each step involved in the process by following the rules of Tower of Hanoi.

Digital Logic Design

1. Convert the following fractional numbers:

(a) Convert 10101.1012 to decimal.

(b) 42.5_{10} to Octal.

2. Subtract 15 and 7 using the binary 2's complement method
3. Convert the following numbers:

(a) 18.75_{10} to Binary

(b) Convert $3F16$ to decimal.

4. Convert 73_{10} into BCD and Excess-3 code.

Numerical Methods

1. Convert the decimal number $(572.625)_{10}$ into binary. Explain each step clearly and verify the result by reverse conversion
2. What are truncation error and round-off error? Compare them and illustrate with examples

3. Perform the following floating-point addition and subtraction (using 4 significant digits and normalized scientific notation): A = 0.725 x 10^2, B = 0.477 x 10^2. Compute A + B and A - B.
4. Write the binary equivalent of the following numbers:
   1. Decimal to Binary: $(43.167)_{10}$ = ?
   2. Octal to Binary: $(37.45)_8$ = ?
5. Define Numerical Computing. Interpret the expression 0.964572E5 + 0.586351E3 and describe the steps required to solve it (no calculation needed).
6. Use the Regula-Falsi method to determine a root of the equation $f(x) = x^3 + 4x - 1$. Accurate to three decimal places.
7. Apply the Newton Raphson method to find a root of the equation $f(x) = x^2 - x - 1$. Correct to three decimal places
8. Using the true value of $\pi = 3.14159265$ and approximate value = 3.14, explain the meaning of absolute error, relative error, and percentage error.
9. Find the significant digits of the following numbers. a) 0.004060 b) 30500 c) 0.0078 d) 12.4500 e) 9.020 x 10^4
10. Explain how loss of significance can affect computational result.

Engineering economics

1. Define market equilibrium. A demand function is given as Qd = (120 - 4p) and a supply function is given as Qs = (2p + 30). Find out equilibrium price and quantity.
2. If both demand and supply increases simultaneously, what may happen to demand and supply and market equilibrium ? Explain.
3. Explain the effect of an increase in consumer income on market equilibrium and the effect of technological improvement on equilibrium.
4. Define Economics. Explain Economic problem of scarcity and choice.
5. Explain how microeconomic theories contribute to efficient resource allocation in an economy.
6. Explain the central problems of an economy.
7. Differentiate between change in demand vs change in quantity demanded with figure.
8. Explain Market Equilibrium with the demand and supply functions created by you. Analyze with figure.
9. A demand and supply function is given as: Qd = (200 - 3P) and Qs = (4P + 80). Show the Adjustment of price and quantity when price becomes high and low. Use figure.

CT-2

1. Explain central problems of an Economy.
2. How these central problems can be solved ?
3. Define demand function. Explain the factors those affect demand function.
4. Draw a demand schedule and demand curve from a demand function created by you.
5. Write down a demand function and supply function. Find out the equilibrium price and quantity.

6. Explain three sector circular flow of income model with diagram.
7. If price of good A increases from Tk 100 to Tk 500, demand of good B decreases from 20 units to 15 units. Find out the value of elasticity and goods type.
8. Proof that, cross elasticity's value ranges from negative to infinity.

Math

1. A leap year contains of 366 days with 29 days in February. If a leap year is selected at random, Compute the probability that the selected leap year will consist of 53 Sundays.
2. In a community there are equal number of males and females. Suppose 5% of the males and 2% of the females are disabled. A person is chosen at random. If this person is male, then find the probability that he is disabled ?
3. In a factory, three machines, M, K and J are used to make biscuits. Machine M makes 20% of the biscuits. Machine J makes 30% of the biscuits. The rest of the biscuits are made by machine K.

   It is known that 7% of the biscuits made by machine J are broken, 5% of the biscuits made by machine K are broken and 3% of the biscuits made by machine M are broken.

   a. Draw a tree diagram to illustrate all the possible outcomes and associated probabilities. A biscuit is selected at random.

   b. Calculate the probability that the biscuit is made by machine K and is not broken.

   c. Calculate the probability that the biscuit is not broken
4. In a community there are males and females are distributed as 2:3 of 100%. Suppose 7% of the males and 12% of the females are disabled. A person is chosen at random. If this person is female, then find the probability that she is not disabled?
5. A pair of dice is thrown. Find the probability that sum of the points on the two dice is 10 or greater if a 5 appears on the first die.
6. In a factory, three machines, J, K and L are used to make biscuits. Machine J makes 25%of the biscuits. Machine K makes 35% of the biscuits. The rest of the biscuits are made by machine L.

   It is known that 5% of the biscuits made by machine J are broken, 6%of the biscuits made by machine K are broken and 2% of the biscuits made by machine L are broken.

   a. Draw a tree diagram to illustrate all the possible outcomes and associated probabilities. A biscuit is selected at random.

   b. Calculate the probability that the biscuit is made by machine J and is not broken.

   c. Calculate the probability that the biscuit is broken.

1. A leap year contains of 366 days with 29 days in February. If a leap year is selected at random, **Compute** the probability that the selected leap year will consist of 53 Mondays. **03**

2. In a community there are males and females are distributed as 2:3 of 100%. Suppose 7% of the males and 12% of the females are disabled. A person is chosen at random. If this person is female, then **find** the probability that she is not disabled? **03**

3. The probability of Saikat going from Dhaka to Barishal by launch is $\frac{5}{7}$ and the probability of going from Barishal to Kuakata by bus is $\frac{3}{5}$. **Find** the probability that Saikat go from Dhaka to Barishal by launch and does not go from Barishal to Kuakata by bus using a probability tree. **04**

4. In a factory, three machines, **J, K and L** are used to make biscuits. Machine **J** makes 25% of the biscuits. Machine **K** makes 35% of the biscuits. The rest of the biscuits are made by machine **L**. It is known that 5% of the biscuits made by machine **J** are broken, 6% of the biscuits made by machine **K** are broken and 2% of the biscuits made by machine **L** are broken. **04**

   a. **Draw** a tree diagram to illustrate all the possible outcomes and associated probabilities. A biscuit is selected at random.
   b. **Calculate** the probability that the biscuit is made by machine **J** and is not broken.
   c. **Calculate** the probability that the biscuit is broken.

1. **Find** the probability of getting a 7 or 11 total on either of two tosses of a pair of fair dice. **03**

2. The probability that a married couple man watches a certain TV show is 0.4 and that his wife watches the show is 0.5. The probability that a man watches the show, given that his wife does is 0.7. **Find** the probability that a wife watches the show given that her husband does. **03**

3. Suppose a sample space consists of 500 persons and are distributed according to their sex and employment status as shown in the accompanying table. **04**

	Employment Status		
Sex	Employed (E)	Unemployed (U)	Total
Male (M)	255	20	275
Female (F)	80	145	225
Total	335	165	500

Find the following probabilities:
   a. the selected person is a male
   b. the selected person is male and female
   c. the selected person is male and unemployed
   d. female and not unemployed

4. In a factory, three machines, **M, K and J** are used to make biscuits. Machine **M** makes 20% of the biscuits. Machine **J** makes 30% of the biscuits. The rest of the biscuits are made by machine **K**. **04**
It is known that 7% of the biscuits made by machine **J** are broken, 5% of the biscuits made by machine **K** are broken and 3% of the biscuits made by machine **M** are broken.

   a. **Draw** a tree diagram to illustrate all the possible outcomes and associated probabilities. A biscuit is selected at random.
   b. **Calculate** the probability that the biscuit is made by machine **K** and is not broken.
   c. **Calculate** the probability that the biscuit is not broken.

7. A leap year contains of 366 days with 29 days in February. If a leap year is selected at random, Compute the probability that the selected leap year will consist of 53 Saturdays.
8. In a community there are equal number of males and females. Suppose 5% of the males and 2% of the females are disabled. A person is chosen at random. If this person is male, then find the probability that he is disabled
9. Turi will go from Dhaka to Rajshahi and then from Rajshahi to Khulna. However, she couldn't decide whether to go by bus or train. So, she tossed a coin twice to check her probability. Finally,it was seen that her probability of going from Dhaka to Rajshahi by bus is $\frac{2}{7}$ and the probability of going from Rajshahi to Khulna by train is $\frac{5}{8}$. Using a Probability Tree, find the probability that Turi does not go to Rajshahi by bus and does not go to Khulna by train.

10. In a factory, three machines, J, K and L are used to make biscuits. Machine J makes 25%of the biscuits. Machine K makes 35% of the biscuits. The rest of the biscuits are made by machine L. It is known that 5% of the biscuits made by machine J are broken, 6%of the biscuits made by machine K are broken and 2% of the biscuits made by machine L are broken.

a. Draw a tree diagram to illustrate all the possible outcomes and associated probabilities. a biscuit is selected at random.

b. Calculate the probability that the biscuit is made by machine J and is not broken.

c. Calculate the probability that the biscuit is broken.

1.	**Compute** the arithmetic mean and geometric mean from the following set of observations: 2, 4, 8, 12, 16, 24.	05
2.	In the winter season, the temperature (in Celsius) of the month of March in Cumilla is placed below. **Construct** the frequency distribution table of the temperature: 10°,6°,13°,13°,14°,6°,10°,10°,11°,12°,11°,10°,9°,8°,9°,11°,12°,10,7°,9°,7°,6°,6°,14°,6°,7°,8°,9°,9°,8°,13°	05

1.	In the winter season, the temperature (in Celsius) of the month of January in BAIUST campus is placed below. **Construct** the frequency distribution table of the temperature: 13°,14°,13°,13°,12°,13°,9°,10°,11°,12°,11°,10°,9°,8°,9°,11°,10°,10,8°,10°,7°,6°,6°,6°,6°,7°,8°,9°,9°,7°,8°	05
2.	The following data refers to the number of children per family of 75 workers of a factory. **Compute** the average number of children per family of the workers of the factory:	05

Number of children	0	1	2	3	4	5	6
Number of families	3	5	10	16	25	12	4

1. **Compute** the geometric mean of the following data: 05

x	10	20	30	40	50	60
f	12	15	25	10	6	2

2. In the winter season, the temperature (in Celsius) of the month of February in Cumilla is 05

1. In the winter season, the temperature (in Celsius) of the month of January in BAIUST campus is placed below. **Construct** the frequency distribution table of the temperature: 05
14°,14°,14°,13°,12°,13°,10°,10°,11°,12°,11°,10°,9°,8°,9°,11°,10°,10,8°,9°,7°,6°,6°,6°,6°,7°,8°,9°,9°,8°,7°

2. The following data refers to the number of years worked by 37 employees of a firm. **Compute** the average number of years worked by the employees of the firm: 05

Number of years worked	5–10	10–15	15–20	20–25	25–30	30–35
Number of employees	5	7	11	8	4	2

1.	The following data refers to the number of children per family of 75 workers of a factory. Compute the average number of children per family of the workers of the factory:	05

Number of children	0	1	2	3	4	5	6
Number of families	3	5	10	16	25	12	4

2.	In the winter season, the temperature (in Celsius) of the month of February in Cumilla is placed below. Construct the frequency distribution table of the temperature:	05

1.	Calculate the arithmetic mean and geometric mean from the following set of observations: 2, 4, 8, 12, 16, 24.	05
2.	The following data refers to the number of years worked by 37 employees of a firm. Compute the average number of years worked by the employees of the firm:	05

Number of years worked	5–10	10–15	15–20	20–25	25–30	30–35
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